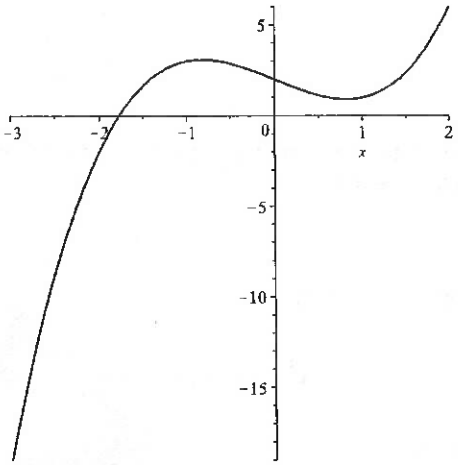
OPTIMISATION

1. a) The graph of the function is in the figure below. Since the function is negative at -2 and positive at -1 and it is continuous it has to change sign, hence it has a zero in the interval $[-2,-1]$.

This is a direct application of the so-called Bolzano Theorem, a special instance of the Intermediate Value Theorem.



[2 marks]

b) The Newton's iteration for the solution of the equation $f(x)=0$ is

$$x_{k+1} = x_k - \frac{f(x_k)}{\frac{\partial f}{\partial x}(x_k)} = \frac{2x_k^3 - 2}{3x_k^2 - 2}.$$

[2 marks]

c) Setting $x_0 = 0$ yields $x_1 = 1$, $x_2 = 0$, $x_3 = 1, \cdots$, hence

$$x_{2k} = x_{2k+2} = 0, \qquad x_{2k+1} = x_{2k+3} = 1,$$

for all $k \geq 0$, which indicates that the sequence $\{x_k\}$ does not converge.

[2 marks]

d) Note that

$$x_{k+2} = x_{k+1} - \frac{f(x_{k+1})}{\frac{\partial f}{\partial x}(x_{k+1})} = \frac{2x_{k+1}^3 - 2}{3x_{k+1}^2 - 2},$$

hence

$$x_{k+2} = \frac{2\dfrac{(2x_k^3-2)^3}{(3x_k^2-2)^3} - 2}{3\dfrac{(2x_k^3-2)^2}{(3x_k^2-2)^2} - 2} = F(x_k).$$

A direct substitution yields $F(0) = 0$ and $F(1) = 1$ which indicates that if, for some $k \geq 0$ one has $x_k = 0$ or $x_k = 1$, then x_{k+2} takes the same value. This explains the periodic behaviour observed in part c). [6 marks]

Answers

e) Consider the quadratic equation $x^2 - c = 0$, with $c > 0$. Note that any quadratic equation of the form $x^2 + bx + c$, with $b^2 - 4ac > 0$, could be written in this form after a change of variable.

i) The Newton's iteration for the solution of the equation $x^2 - c = 0$ is

$$x_{k+1} = \frac{x_k^2 + c}{2x_k}.$$

[2 marks]

ii) Similarly to what done in part d) one obtains

$$x_{k+2} = \frac{\frac{(x_k^2+c)^2}{4x_k^2} + c}{\frac{x_k^2+c}{x_k}} = G(x_k).$$

To determine the fixed points of G one has to solve the equation $x = G(x)$, that is

$$x^2 + c = \frac{(x^2+c)^2}{4x^2} + c$$

or, equivalently,

$$x^3 + cx = cx + \frac{(x^2+c)^2}{4x} \Leftrightarrow x^3 = \frac{(x^2+c)^2}{4x}$$

hence

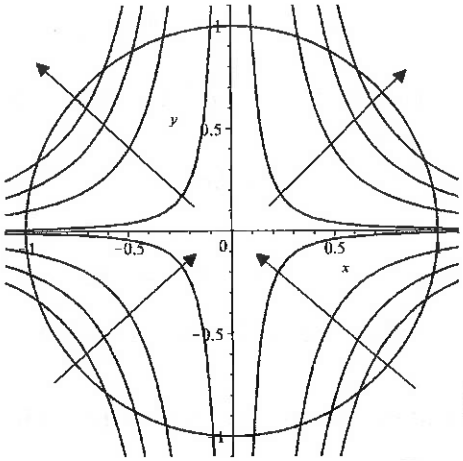
$$(2x^2)^2 = (x^2+c)^2,$$

which has only two real solutions: $x = \pm\sqrt{c}$. This implies that the periodic behaviour observed for the cubic equation cannot occur for the quadratic equation, that is period-two oscillations cannot occur.

[6 marks]

Answers

2. a) The admissible set and the level lines are indicated in the figure below. Note that the value of the function f increases along the arrows and the function is negative for negative y and positive for positive y. A direct inspection shows that all points of the admissible set are regular points for the constraint. Alternatively, it is easy to check that the gradient of g is no-where zero in the admissible set.



[2 marks]

b) The Lagrangian for the problem is

$$L(x,y,\lambda) = x^2 y + \lambda(x^2+y^2-1).$$

The necessary conditions of optimality are

$$0 = \frac{\partial L}{\partial x} = 2\lambda x + 2xy, \qquad 0 = \frac{\partial L}{\partial y} = 2\lambda y + x^2, \qquad x^2+y^2-1 = 0.$$

To determine candidate optimal solutions one could proceed as follows. From the first equation one could consider two cases: $x = 0$ or $\lambda + y = 0$.

In the first case one has, using the constraint, $y = \pm 1$ and, using the second equation, $\lambda = 0$, yielding the two candidate solutions

$$S_1 = \{x = 0, y = 1, \lambda = 0\}, \qquad S_2 = \{x = 0, y = -1, \lambda = 0\}.$$

In the second case one obtains $y + \lambda = 0$, which replaced in the second and third equations yields

$$-2\lambda^2 + x^2 = 0, \qquad \lambda^2 + x^2 = 1.$$

One can easily solve these two equations obtaining four candidate solutions

$$S_3 = \left\{x = \frac{\sqrt{6}}{3}, y = \frac{\sqrt{3}}{3}, \lambda = -\frac{\sqrt{3}}{3}\right\} \qquad S_4 = \left\{x = -\frac{\sqrt{6}}{3}, y = \frac{\sqrt{3}}{3}, \lambda = -\frac{\sqrt{3}}{3}\right\}$$

$$S_5 = \left\{x = \frac{\sqrt{6}}{3}, y = -\frac{\sqrt{3}}{3}, \lambda = \frac{\sqrt{3}}{3}\right\} \qquad S_6 = \left\{x = -\frac{\sqrt{6}}{3}, y = -\frac{\sqrt{3}}{3}, \lambda = \frac{\sqrt{3}}{3}\right\}$$

[6 marks]

Answers

c) Note that

$$\nabla^2 L = 2\begin{bmatrix} y+\lambda & x \\ x & \lambda \end{bmatrix}$$

and that $\dfrac{\partial g}{\partial(x,y)} = 2\begin{bmatrix} x & y \end{bmatrix}$. Thus, $\dfrac{\partial g}{\partial(x,y)}s = 0$ yields

$$s = \alpha\begin{bmatrix} -y \\ x \end{bmatrix},$$

with $\alpha \neq 0$, hence $s^T\nabla^2 Ls = 8\alpha^2(\lambda x^2 + \lambda y^2 - 2x^2 y + y^3)$. As a result,

$$s^T\nabla^2 Ls(S_1) = 8\alpha^2, \qquad s^T\nabla^2 Ls(S_5) = 16\frac{\sqrt{3}}{3}\alpha^2, \qquad s^T\nabla^2 Ls(S_6) = 16\frac{\sqrt{3}}{3}\alpha^2,$$

$$s^T\nabla^2 Ls(S_2) = -8\alpha^2, \quad s^T\nabla^2 Ls(S_3) = -16\frac{\sqrt{3}}{3}\alpha^2, \quad s^T\nabla^2 Ls(S_4) = -16\frac{\sqrt{3}}{3}\alpha^2,$$

hence S_1, S_5, and S_6 are local minimizers and S_2, S_3, and S_4 are local maximizers. [4 marks]

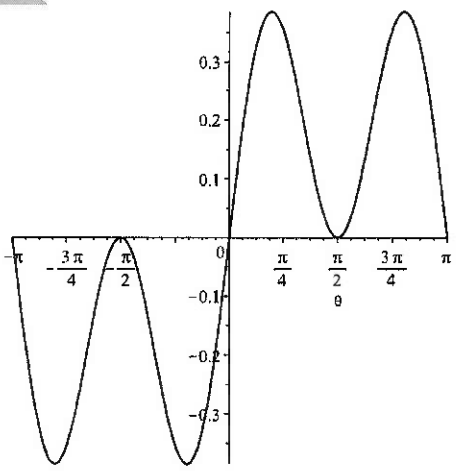
d) The values of the objective function at the minimizers and maximazers are as follows

$$f(S_1) = 0, \qquad f(S_5) = f(S_6) = -2\frac{\sqrt{3}}{9}$$

$$f(S_2) = 0, \qquad f(S_3) = f(S_4) = 2\frac{\sqrt{3}}{9}$$

These values indicate that $f(S_1) = f(S_2)$, that is the value of the objective function at a local minimizer and at a local maximer is the same, that S_5 and S_6 are global minimizers and that S_3 and S_4 are global maximizers. [4 marks]

e) Using the indicated change of variables one has $f(cos\theta, \sin\theta) = \cos^2\theta \sin\theta$, and $g(\cos\theta, \sin\theta) = \cos^2\theta + \sin^2\theta = 1$, which shows that the problem has been transformed into an unconstrained problem, since the equality constraint is trivially satisfied for all θ, with objective function $f(\theta)$. The graph of f is in the figure below, which shows that the function has three minimizers/maximizers, that there are two global minimizers/maximizers, and that the value of the function at a local minimizer equals the value of the function at a local maximizer, as per the conclusions in parts c) and d).



[4 marks]

Answers

3. a) The Lagrangian of the problem is (note the change in sign to transform the problem into a minimization problem)

$$L(r_1, r_2, r_3, \rho) = -\ln r_1 - 2\ln r_2 - 3\ln r_3 + \rho(r_1 + r_2 + r_3 - C).$$

The first order necessary conditions of optimality are

$$0 = \frac{\partial L}{\partial r_1} = -\frac{1}{r_1} + \rho, \qquad 0 = \frac{\partial L}{\partial r_2} = -\frac{2}{r_2} + \rho, \qquad 0 = \frac{\partial L}{\partial r_3} = -\frac{3}{r_3} + \rho,$$

$$r_1 + r_2 + r_3 - C \leq 0, \qquad \rho \geq 0, \qquad \rho(r_1 + r_2 + r_3 - C) = 0.$$

[2 marks]

b) Using the complementarity conditions one has two cases: $\rho = 0$ and $r_1 + r_2 + r_3 - C = 0$. The former does not yield any candidate solution, since the equations $0 = \frac{\partial L}{\partial r_i}$ do not have any solution for $\rho = 0$, hence one has to consider only the latter, which in particular implies that the inequality constrain is satisfied with the equality sign at any candidate optimal solution. Solving the four equations

$$0 = \frac{\partial L}{\partial r_1} = \frac{\partial L}{\partial r_2} = \frac{\partial L}{\partial r_3} = r_1 + r_2 + r_3 - C$$

yields the candidate solution

$$r_1 = \frac{C}{6}, \qquad r_2 = \frac{C}{3}, \qquad r_3 = \frac{C}{2}, \qquad \rho = \frac{6}{C}.$$

Note that the candidate optimal r_i's are positive (recall that $C > 0$). [4 marks]

c) A direct substitution yields

$$\begin{aligned} S(r, \lambda) \;=\; & -\ln r_1 - 2\ln r_2 - 3\ln r_3 + \lambda(r_1 + r_2 + r_3 - C) + \\ & \frac{1}{\varepsilon}(r_1 + r_2 + r_3 - C)^2 + \eta\left(-\frac{1}{r_1} - \frac{2}{r_2} - \frac{3}{r_3} + 3\lambda\right)^2. \end{aligned}$$

Computing the gradient of S with respect to r_1, r_2, r_3, and λ yields

$$\frac{\partial S}{\partial r_1} = -\frac{1}{r_1} + \rho + \frac{2}{\varepsilon}(r_1 + r_2 + r_3 - C) + \frac{2\eta}{r_1^2}\left(-\frac{1}{r_1} - \frac{2}{r_2} - \frac{3}{r_3} + 3\rho\right),$$

$$\frac{\partial S}{\partial r_2} = -\frac{2}{r_2} + \rho + \frac{2}{\varepsilon}(r_1 + r_2 + r_3 - C) + \frac{4\eta}{r_2^2}\left(-\frac{1}{r_1} - \frac{2}{r_2} - \frac{3}{r_3} + 3\rho\right),$$

$$\frac{\partial S}{\partial r_3} = -\frac{3}{r_3} + \rho + \frac{2}{\varepsilon}(r_1 + r_2 + r_3 - C) + \frac{6\eta}{r_3^2}\left(-\frac{1}{r_1} - \frac{2}{r_2} - \frac{3}{r_3} + 3\rho\right),$$

$$\frac{\partial S}{\partial \lambda} = (r_1 + r_2 + r_3 - C) + 6\eta\left(-\frac{1}{r_1} - \frac{2}{r_2} - \frac{3}{r_3} + 3\rho\right).$$

Note that the candidate optimal point is such that $r_1 + r_2 + r_3 - C = 0$, that is it satisfies the inequality constraint with the equality sign, and it is such that $-\frac{1}{r_1} - \frac{2}{r_2} - \frac{3}{r_3} + 3\rho = 0$, that is all terms with ε and η are zero. Finally, the remaining terms are also zero once evaluated at the candidate optimal point.

[6 marks]

Answers

d) The Lagrangian of the problem is (note again the change in sign to transform the problem into a minimization problem)

$$L(r_1, r_2, \cdots, r_n, \rho) = -\alpha_1 \ln r_1 - \alpha_2 \ln r_2 - \cdots - \alpha_n \ln r_n + \rho(r_1 + r_2 + \cdots r_n - C).$$

The first order necessary conditions of optimality are such that

$$0 = \frac{\partial L}{\partial r_i} = -\alpha_i \frac{1}{r_i} + \rho,$$

yielding

$$r_i = \frac{\alpha_i}{\rho}.$$

Exploiting the results of part b) we can conclude that the inequality constraint is satisfied with the equality sign, hence

$$r_1 + r_2 + \cdots + r_n = \frac{\alpha_1}{\rho} + \frac{\alpha_2}{\rho} + \cdots + \frac{\alpha_n}{\rho} = C.$$

This yields

$$\rho = \frac{\alpha_1 + \alpha_2 + \cdots + \alpha_n}{C}$$

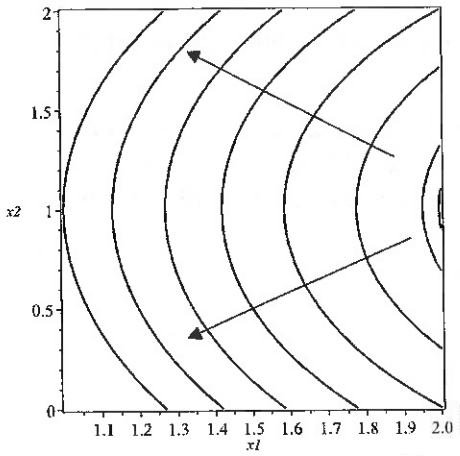
and the candidate solutions

$$r_i = C \frac{\alpha_i}{\alpha_1 + \alpha_2 + \cdots + \alpha_n},$$

which satisfy the necessary conditions of optimality and the constraint. [8 marks]

Answers

4. a) The level lines of the cost function, which are circles centered at $(3,1)$ and the admissible set are displayed in the figure below. Note that the value of the function f increases along the arrows and the admissible set is the square bounding the figure. The function f is increasing along the direction of the arrows. The optimal solution is the point $(2,1)$.



[2 marks]

b) i) The linear problem in (S1) is

$$\min_x \; x_1(2x_1^k - 6) + x_2(2x_2^k - 2), \qquad Ax \leq b,$$

with

$$A = \begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix}.$$

For $k = 0$ and $x^0 = (0,0)$ one has $\min_x \; -6x_1 - 2x_2$, subject to the constrains $Ax \leq b$. The optimal solution is therefore given by the largest feasible x_1 and x_2, that is $\xi = (2,2)$. [4 marks]

ii) A direct calculation yields $(\xi - x^0)^T \nabla f(x^0) = -16 < 0$. The direction $\xi - x^0$ is given by the vector $v = \begin{bmatrix} 2 & 2 \end{bmatrix}^T$, that is

$$x^1 = x^0 + t_0 v = \begin{bmatrix} 2t_0 \\ 2t_0 \end{bmatrix}.$$

Evaluating the function f at x^1 yields $f(2t_0, 2t_0) = 8t_0^2 - 16t_0 + 10$. To perform an exact line search one has to minimize this function with respect to t_0, yielding the optimal $t_0 = 1$ and, hence $x^1 = (2,2)$.

[4 marks]

c) One has to repeat the steps of part b) from x^1.

i) The linear problem in (S1) is now $\min_x \; -2x_1 + 2x_2$, subject to the constrains $Ax \leq b$. The optimal solution is therefore given by the largest feasible x_1 and the smallest feasible x_2, that is $\xi = (2,0)$. [4 marks]

ii) A direct calculation yields $(\xi - x^1)^T \nabla f(x^1) = -4 < 0$. Note that the optimality gap has been reduced. The direction $\xi - x^1$ is given by the

Answers

vector $v = \begin{bmatrix} 0 & -2 \end{bmatrix}^T$, that is

$$x^2 = x^1 + t_1 v = \begin{bmatrix} 2 \\ 2 - 2t_1 \end{bmatrix}.$$

Evaluating the function f at x^2 yields $f(2, 2-2t_1) = 4t_1^2 - 4t_1 + 2$. To perform an exact line search one has to minimize this function with respect to t_1, yielding the optimal $t_1 = 1/2$ and, hence $x^2 = (2, 1)$.

[4 marks]

d) A direct substitution of the indicated value of ξ and of the value of x^2 obtained in part c) yields that the optimality gap $(\xi - x^2)^T \nabla f(x_2) = 0$, hence x_2 is the optimal point. This coincides with the optimal solution determined in part a).

[2 marks]

Answers